higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANOTECHNICS N5

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This marking guideline consists of 8 pages.

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QUESTION 1

1.1
- Spur gears
- Helical gears
- Double-helical gears (3)

1.2 1.2.1

CONDITION	GEAR A	GEAR B	GEAR C	ARM D	ARM E
Fix arm D and arm E rotate Gear A by +1 rev	$+1\surd$	$=+\frac{40}{32}$ $=1,25\surd$	$=\frac{40}{32}x\frac{32}{-24}$ $-1,667\surd$	$0\surd$	0
Multiply by x and add y	$x+y\surd$	$1,25x+y$ $\surd$	$-1,667x+y$ $\surd$	$+y\surd$	$+y$
$N_D=+120r/\min$ $N_E=+120r/\min$ $N_A=0$ $N_B=?$	$N_A=0\surd$	$N_B=?\surd$		$N_D=+120r/\min$ $\surd$	$N_E=+120r/\min$
$N_D=+120r/\min$ $N_E=+120r/\min$ $N_A=0$ $N_C=?$	$N_A=0\surd$		$N_C=?\surd$	$N_D=+120r/\min$ $\surd$	$N_E=+120r/\min$

(7)

Determine x and y

$y=+120$ ✓

$x+y=0$

$x=-y$ ✓

$N_B=1,25x+y$

$=1,25(-120)+120$ ✓

$=-150+120$

$=-30$

$N_B=-30\ r/\min$ ✓ *Anticlockwise OR Opposite direction*

(4)

1.2.2 *Determine x and y*

$y=+120$ ✓

$x+y=0$

$x=-y$ ✓

$=-1,667(-120)+120$ ✓

$=200,04+120$

$=320,04r/\min$

$N_C=-1,667x+y$

$N_C=320,04\ r/\min$ ✓

Nc rotates 320,04 r/min clockwise or positive direction

(4)
[18]

QUESTION 2

2.1
- Gravel
- Sugar
- Flour
- Fertiliser
- Coke
- Sand
- Lime

(Any 4 × 1) (4)

2.2 2.2.1

$$mass\ of\ load = \frac{400}{3,6} \checkmark$$

$$mass\ of\ load = 111,111\ kg/s \checkmark$$

$$volume\ delivered = \frac{mass\ of\ load}{density\ of\ load}$$

$$= \frac{111,111}{900} \checkmark$$

$$volume\ delivered = 0,123\ m^3/s \checkmark$$

$$number\ of\ bucket/s = \frac{v}{s}$$

$$= \frac{0,4}{1} \checkmark$$

$$number\ of\ bucket/s = 0,4/s \checkmark$$

$$volume\ of\ bucket = \frac{volume\ delivered}{no.of\ bucket/s}$$

$$volume\ of\ bucket = \frac{0,123}{0,4} \checkmark$$

$$volume\ of\ bucket = 0,3075\ m^3 \checkmark$$

(8)

2.2.2

$$P_g = mgh$$

$$P_g = 111,111 \times 9,81 \times 50 \checkmark$$

$$P_g = 54499,946\ W \checkmark$$

$$\eta = \frac{P_o}{P_i} \times 100$$

$$80 = \frac{54499,946}{P_i} \times 100 \checkmark\checkmark$$

$$P_i = 68,125\ kW \checkmark$$

(5)
[17]

QUESTION 3

3.1
- Longer life span
- Heavier load can be transported
- Steeper inclines can be accommodated
- Higher capacities can be handled (4)

3.2 3.2.1

$$For\ {}^{6}\!/_{7}\ construction\ rope\ area = 0{,}36D^2$$

$$Ultimate\ stress = \frac{ultimate\ \ Strength}{rope\ \ area}$$

$$= \frac{ultimate\ \ strength}{0{,}36D^2}$$

$$D^2 = \frac{ultimate\ \ strength}{0{,}36 \times \max\ \ stress} \checkmark$$

$$D^2 = \frac{600 \times 10^3}{0{,}36 \times 1300 \times 10^6} \checkmark$$

$$D = \sqrt{0{,}001282} \checkmark$$

$$D = 0{,}0358\ m$$

$$D = 35{,}805\ mm \checkmark$$

(4)

3.2.2

$$rope\ \ mass/m = \frac{d^2}{250}$$

$$= \frac{(35{,}805)^2}{250} \checkmark$$

$$= 5{,}128\ kg/m \checkmark$$

(2)

3.2.3

$$tension\ \ self\ \ weight = \frac{m_1 \times g \times s^2}{8h}$$

$$= \frac{5{,}128 \times 9{,}81 \times (90)^2}{8 \times 2{,}6} \checkmark$$

$$= 19\ 590{,}193\ N \checkmark$$

(2)

3.2.4

$$tension\ \ due\ \ to\ \ one\ \ load = \frac{m_2 \times g \times s}{4h}$$

$$= \frac{460 \times 9{,}81 \times 90}{4 \times 2{,}6} \checkmark$$

$$= = \frac{406134}{10{,}4}$$

$$= 39051{,}346\ N \checkmark$$

(2)

3.2.5 $$Factor\ of\ safety = \frac{ultimate\ \ strength}{working\ \ strength}$$

$$= \frac{600 \times 10^3}{tension\ \ weight + tension\ \ one\ \ load} \checkmark$$

$$= \frac{600 \times 10^3}{19590{,}193 + 39051{,}346} \checkmark$$

$$= \frac{600 \times 10^3}{58641{,}539} \checkmark$$

$$= 10{,}239 \checkmark$$

(4)
[18]

QUESTION 4

4.1
- The diameter
- The number of strands
- The number of wires per strand
- The number of wires in each layer (starting from outside)

(4)

4.2 $$v = u + at$$

$$6 = 0 + (a \times 8) \checkmark$$

$$6 = 8a$$

$$a = 0{,}75\ m/s^2 \checkmark$$

$$T_1 = m_1(g + a)$$

$$= 1600(9{,}81 + 0{,}75) \checkmark\checkmark$$

$$= 16896\ N \checkmark$$

$$T_2 = m_2(g - a)$$

$$= 500(9{,}81 - 0{,}75) \checkmark\checkmark$$

$$= 4530\ N \checkmark$$

$$T_{load} = (T_1 - T_2)r$$

$$= (16896 - 4530)0{,}5 \checkmark$$

$$= 6183\ N.m \checkmark$$

$$P = T_{total} \times \frac{v}{r}$$

$$P = (T_{load} + T_{rope} + T_u) \times \frac{v}{r}$$

$$= (6183 + 101{,}25 + 900) \times \frac{6}{0{,}5} \checkmark$$

$$= 7184{,}25 \times 12 \checkmark$$

$$= 862111\ W$$

$$= 86{,}211\ kW \checkmark$$

(13)

[17]

QUESTION 5

5.1 5.1.1

$$u = \frac{40 \times 10^3}{3600}$$
$$= 11{,}111\ m/s \checkmark$$

$$v = \frac{100 \times 10^3}{3600}$$
$$= 27{,}778\ m/s \checkmark$$ (2)

5.1.2

$$v = u + at$$
$$27{,}778 = 11{,}111 + a \times 30 \checkmark$$
$$16{,}667 = 30a$$
$$a = 0{,}556\ m/s^2 \checkmark$$ (2)

5.2 5.2.1

$$T_E = F_R$$
$$T_E = 0{,}89\%\ of\ 350 \times 10^3 9{,}81 \checkmark$$
$$= 3055815\ N$$
$$= 3055{,}815\ kN \checkmark$$ (2)

5.2.2

$$F_g = m \times 10^3 \times g \times \frac{1}{130}$$
$$F_g = 350 \times 10^3 \times 9{,}81 \times \frac{1}{130} \checkmark$$
$$= 26\ 411{,}538\ \text{N}$$
$$= 26{,}411\ kN \checkmark$$ (2)

5.2.3

$$F_a = m \times a$$
$$F_a = 350 \times 10^3 \times 0{,}09 \checkmark$$
$$= 31\ 500\ \text{N}$$
$$= 31{,}5\ kN \checkmark$$ (2)

5.2.4

$$T_E = F_{R_{total}} + F_{g_{total}} + F_{a_{total}}$$
$$= 3055815 + 26411 + 31500 \checkmark$$
$$= 3\ 113\ 726\ \text{N} \checkmark$$

$$P_{loco} = T_E \times v$$
$$= 3113726 \times \frac{64 \times 10^3}{3\ 600} \checkmark$$
$$= 55355128{,}889\ W$$
$$= 55{,}355\ MW \checkmark$$ (4)

5.2.5

$$E_K = \frac{1}{2} m \times v^2$$
$$= \frac{1}{2} \times 350 \times 10^3 \times (17,778)^2 \checkmark$$
$$= 55,310\ MJ \checkmark$$

(2)
[16]

QUESTION 6

6.1

$$\omega_1 = \frac{2\pi N}{60}$$
$$= \frac{2 \times \pi \times 100}{60} \checkmark$$
$$= 10,472\ rad/s \checkmark$$

$$\omega_2 = \frac{2\pi N}{60}$$
$$= \frac{2 \times \pi \times 130}{60} \checkmark$$
$$= 13,614\ rad/s \checkmark$$

$$\alpha = \frac{\omega_2 - \omega_1}{t}$$
$$= \frac{13,614 - 10,472}{35} \checkmark$$
$$= 0,089\ rad/s^2 \checkmark$$

$$T = m.k^2.\alpha$$
$$= 1400 \times (0,82)^2 \times 0,089 \checkmark$$
$$= 83,781\ N.m \checkmark$$

(8)

6.2 6.2.1

$$\omega = \frac{2\pi N}{60}$$
$$= \frac{2x\pi x95}{60}$$
$$= 9,948 rad/s$$

$$E_k = \frac{1}{2} m.k^2.\omega^2$$
$$= \frac{1}{2} x600x(0,83)^2 x(9,948)^2 \checkmark$$
$$= 20\ 459,476\ \text{J}$$
$$= 20,459\ \text{kJ} \checkmark$$

(2)

6.2.2 $E_k = \theta.T$

$$T = \frac{E_k}{\theta}$$

$$= \frac{20456}{376,991} \checkmark$$

$$= 54,261\ Nm \checkmark$$

(2)

6.2.3 $T = m.k^2.\alpha$

$$\alpha = \frac{T}{m.k^2}$$

$$= \frac{54,261}{600x(0,83)^2}$$

$$= 0,131 rad/s^2 \checkmark$$

$$\omega_2 = \omega_1 + \alpha.t$$

$$t = \frac{\omega_2 - \omega_1}{\alpha}$$

$$= \frac{11,309 - 0}{0,131}$$

$$= 86,328\ s$$

$$say = 86\ s \checkmark$$

(2)
[14]

TOTAL: 100

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